LOMITHA WICKRAMARACHCHI

Final Year Electrical and Electronic Engineering Student | Power Systems and Grid Connection
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PROFILE

Final year Electrical and Electronic Engineering (Honours) student at Swinburne University of Technology, graduating December 2026 and specialising in power systems and renewable energy. As a Research Assistant at the Siemens Energy Hub, I build and validate electromagnetic transient (EMT) inverter models in PSCAD/EMTDC for an industry client. I also bring HV field experience on AusNet and Jemena networks and co-authored an IEEE conference paper on vehicle to grid support and active filtering. I am seeking a graduate role in grid connection, power system studies or network engineering.

EDUCATION

Bachelor of Engineering (Honours), Electrical and Electronic 2022 to Dec 2026

Swinburne University of Technology, Melbourne | Specialisation: power systems and renewable energy

- **Relevant units:** Electrical Power Systems, Power Protection, Power Systems Operation and Control, Power Electronics, Hydrogen and Energy Storage, Design of Smart Power Grids (in progress)
- **Training:** PSS/E Modules 1 and 2 and PSS SINCAL (June 2026); VESI accreditation (Nov 2025)

ENGINEERING EXPERIENCE

Research Assistant May 2026 to Present

Siemens Energy Hub, Swinburne University of Technology

- Lead the conversion of a FIMER central solar inverter's control from a sealed, compiled DLL into a fully native PSCAD/EMTDC model, built from PSCAD blocks and Fortran, that drops into the manufacturer's own test case.
- Rebuilt the fault ride-through and Q(U) reactive power subsystems from MATLAB/Simulink as native PSCAD blocks, matching the Simulink reference at every output before integration.
- Delivered 59 controlled model changes, each passed through offline reference checks, a clean build, bit identity regression and a side by side run against the vendor's model; the native model reproduces the manufacturer's results across dips, swells, a phase jump, a frequency step and an LVRT trip.
- Model covers PLLs, inner current control, P(f) and under frequency control, fault ride-through with negative sequence current support, and LVRT, HVRT and frequency protection.
- Traced client reported faults to test bench wiring, time step artefacts and block execution order; supplied corrected test cases and kept a written change log for every model revision.
- Main technical contact with the client: delivered model packages and presented at review meetings. Scripted PSCAD builds and test runs in Python through the PSCAD automation interface, using AI assisted development tools.

Electrical Engineering Intern Nov 2025 to Jan 2026

Spring Power Solutions Pty Ltd, Hallam VIC

- Assisted senior engineers with HV diagnostics and pre-commissioning testing of assets on the AusNet and Jemena networks.
- Performed VLF withstand testing, partial discharge and tan delta diagnostics on MV cables at kiosk and substation sites, and analysed transformer test reports.
- Worked under permits, isolations and controlled discharge procedures on network infrastructure; completed VESI accreditation and AusNet site induction.

PUBLICATIONS

[1] N. S. Mathew, A. L. Wickramarachchi, et al., "Autonomous fuzzy energy management for vehicle-to-grid support in a bidirectional EV fast-charging system," accepted for presentation at the IEEE Conf. Engineering Informatics (ICEI 2026).

[2] N. S. Mathew, A. L. Wickramarachchi, I. Kavalchuk, G. Thirunavukkarasu, K. Ahmed and M. Seyedmahmoudian, "Active filtering and autonomous V2G dispatch for EV fast-charging sites," journal manuscript, 2026.

ENGINEERING PROJECTS

More detail, figures and results for each project are on my portfolio: lomi-07.github.io/Lomitha_portfolio

Capstone: Active Filtering and Autonomous V2G Dispatch for EV Fast Charging Sites 2026

- Modelled a 400 V site with a bidirectional EV charger, a shunt active power filter and a 29 kW six pulse rectifier load as a switched model in MATLAB/Simulink and Simscape Electrical.
- Built the active front end's dq frame control (SRF-PLL, decoupled current loops, 10 kHz SPWM), the bidirectional DC/DC stage and the battery model.

- Filter cut grid current TDD from 15.4 % to 2.75 %, inside the 8 % IEEE Std 519-2022 limit; a fuzzy energy management system reversed battery power through a 0.85 pu sag with the 750 V DC link held within +6.25 % and −5.66 %.
- Used small signal analysis to trace a 24 % current overshoot to the battery current PI controller and showed the supervisory loop gain stays below 0.2.

Dynamic Operating Envelopes Under Bad Data (EEE40016, team project) 2026

- Studied how telemetry latency, packet loss and battery SoC error affect dynamic operating envelope control against a static EV charging limit, using Python control logic and PSS SINCAL on the IEEE European LV Test Feeder; owned the EV charge point controller node.

AEMO DMAT Test Automation (personal project, in progress) 2026

- Building a Python harness to automate AEMO Dynamic Model Acceptance Tests on inverter models in PSS/E, with test definitions traced to the AEMO guideline.

Earlier projects: SDI-12 environmental data logger (team lead), LabVIEW railway level crossing, MQTT real time sensor monitoring.

TECHNICAL SKILLS

Power system simulation: PSCAD/EMTDC, PSS/E, PSS SINCAL, MATLAB/Simulink, Simscape Electrical, PSIM

Programming: Python (numpy, pandas, matplotlib, PSCAD automation), Fortran, MATLAB, C/C++

Grid codes and standards: AEMO DMAT, IEEE Std 519-2022, IEEE Std 1547, AS/NZS 4777.2

Testing and field: VLF withstand, partial discharge, tan delta, transformer test analysis, permits and isolations

AI assisted engineering: Claude Code and AI agents to automate simulation builds, test runs, results analysis and reporting, with every output reviewed and validated against a reference

Automation and tools: NI LabVIEW, Codesys, Arduino, Git, Microsoft Office

OTHER EXPERIENCE

Safety and Security Team Leader 2024 to Present

Arts Centre Melbourne and Sidney Myer Music Bowl (through Advent Security)

- Monitor more than 700 CCTV cameras, 300 access control points and fire and gas alarm systems across a major performing arts venue.
- Coordinate fire system isolations, respond to technical system alerts and lead security teams through high pressure events.

Volunteering and Coaching 2020 to 2025

- Swinburne certified volunteer (100+ hours, 2024 to 2025): English tutor for exchange students, orientation peer mentor and campus event support.
- Coached an under 10 team at Colombo Kickers Football Academy (2020 to 2021) and supported the under 18 squad at an international tournament in Vietnam.

REFERENCES

Available on request.